

Exercise = 9.1

Q No: 01

Verify Euler's Theorem for

$$a: u = \arcsin\left(\frac{x}{y}\right) + \arctan\left(\frac{y}{x}\right)$$

Sol:

$$\begin{aligned} u &= \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) \\ &= \sin^{-1}\left(\frac{tx}{ty}\right) + \tan^{-1}\left(\frac{ty}{tx}\right) \end{aligned}$$

Multiply by t to prove homogenous function.

$$\begin{aligned} &= 1 \cdot \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) & \because 1 = t^0 \\ &= t^0 \left[\sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) \right] \end{aligned}$$

$$\therefore n = 0$$

This show given "U" is homogenous function. & its degree is $n=0$.
So Euler's theorem is:

$$\Rightarrow x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = 0 \quad \text{--- ①}$$

Partial diff ① w.r.t "x"

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x} \left[\sin^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y}{x} \right) \right]$$

$$= \frac{1}{\sqrt{1 - \left(\frac{x}{y} \right)^2}} \cdot \frac{\partial}{\partial x} \left(\frac{x}{y} \right) + \frac{1}{1 + \left(\frac{y}{x} \right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot \frac{1}{y} \frac{\partial}{\partial x} (x) + \frac{1}{1 + \frac{y^2}{x^2}} \cdot y \cdot \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$= \frac{1}{\sqrt{\frac{y^2 - x^2}{y^2}}} \cdot \frac{1}{y} (1) + \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot y \left[\frac{-1}{x^2} \right]$$

$$= \frac{y}{\sqrt{y^2 - x^2}} \cdot \frac{1}{y} + \frac{x^2}{x^2 + y^2} \cdot \frac{y}{x^2}$$

$$\frac{\partial U}{\partial x} = \frac{1}{\sqrt{y^2 - x^2}} - \frac{y}{x^2 + y^2} \quad \text{--- ②}$$

Partial diff ① w.r.t "y"

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left[\sin^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y}{x} \right) \right]$$

$$= \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \cdot x \frac{\partial}{\partial y} + \frac{1}{1+\frac{y^2}{x^2}} \cdot \frac{1}{x} \frac{\partial}{\partial y} (y) \quad \begin{matrix} \therefore \sin^{-1} = \sqrt{1-x^2} \\ \therefore \tan^{-1} = 1+x^2 \end{matrix}$$

$$= \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \cdot x \left(-\frac{1}{y^2} \right) + \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{1}{x} (1)$$

$$= \frac{1}{\frac{\sqrt{y^2-x^2}}{\sqrt{y^2}}} \cdot x \left(-\frac{1}{y^2} \right) + \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{1}{x} (1)$$

$$= \frac{y}{\sqrt{y^2-x^2}} \cdot \frac{-x}{y^2} + \frac{x^2}{x^2+y^2} \cdot \frac{1}{x}$$

$$\frac{\partial v}{\partial y} = \frac{-x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2} \quad \text{--- (3)}$$

Put in formula equation (2) and (3)

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial v}{\partial x} = \frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2}$$

$$\frac{\partial v}{\partial y} = \frac{-x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2}$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0$$

$$x \left(\frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2} \right) + y \left(\frac{-x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2} \right) = 0$$

$$\frac{x}{\sqrt{y^2-x^2}} - \frac{xy}{x^2+y^2} - \frac{x}{\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2} = 0$$

$$0 = 0$$

$$L.H.S = R.H.S$$

Hence proved Euler's Theorem.

b. $u = x^n \cdot \ln \left[\frac{y}{x} \right]$

Sol:

$$u = x^n \cdot \ln \left[\frac{y}{x} \right] \quad \text{--- (1)}$$

By checking homogenous function multiply t on function.

$$\begin{aligned} U &= (tx)^n \cdot \ln \left[\frac{ty}{tx} \right] \\ &= t^n \cdot \left[x^n \cdot \ln \left[\frac{y}{x} \right] \right] \end{aligned}$$

It is homogenous function with n th degree.

This show "U" is homogenous function with degree "n". So, Euler's theorem is:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad \text{--- (2)}$$

Take partial diff (1) w.r.t "x".

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[x^n \cdot \ln \left(\frac{y}{x} \right) \right]$$

$$= x^n \cdot \frac{\partial}{\partial x} \ln\left(\frac{y}{x}\right) + \ln\left(\frac{y}{x}\right) \cdot \frac{\partial}{\partial x} (x^n)$$

$$= x^n \cdot \frac{1}{\frac{y}{x}} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x}\right) + \ln\left(\frac{y}{x}\right) \cdot n \cdot x^{n-1}$$

$$= x^n \cdot \frac{x}{y} \cdot y \frac{\partial}{\partial x} \left(\frac{1}{x}\right) + \ln\left(\frac{y}{x}\right) \cdot n \cdot x^{n-1}$$

$$= x^n \cdot x \cdot \left(-\frac{1}{x^2}\right) + n \cdot x^{n-1} \cdot \ln\left(\frac{y}{x}\right)$$

$$\frac{\partial v}{\partial x} = -\frac{x^n}{x} + n \cdot x^{n-1} \ln\left(\frac{y}{x}\right)$$

Partial diff ① w.r.t "y"

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left[x^n \cdot \ln\left(\frac{y}{x}\right) \right]$$

$$= x^n \cdot \frac{\partial}{\partial y} \ln\left(\frac{y}{x}\right)$$

$$= x^n \cdot \frac{1}{\frac{y}{x}} \cdot \frac{\partial}{\partial y} \left(\frac{y}{x}\right)$$

$$\therefore \left(\frac{y}{x}\right) = y \cdot x^{-1}$$

$$= x^n \cdot \frac{x}{y} \cdot \frac{1}{x} \cdot \frac{\partial}{\partial y} (y)$$

$$= x^n = \frac{x^n}{y}$$

$$\frac{\partial v}{\partial y} = \frac{x^n}{y}$$

Put in equation ①

$$x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = nU$$

$$x \left[\frac{-x^n}{x} + nx^{n-1} \ln \left(\frac{y}{x} \right) \right] + y \left[\frac{x^n}{y} \right] = nU$$

$$-x^n + x \cdot nx^{n-1} \cdot \ln \left(\frac{y}{x} \right) + x^n = nU$$

$$-x^n + nx^n \cdot \ln \left(\frac{y}{x} \right) + x^n = nU$$

$$n \cdot x^n \ln \left(\frac{y}{x} \right) = nU$$

$$\therefore U = x^n \ln \left(\frac{y}{x} \right)$$

$$n \cdot U = nU$$

$$L: H: S = R: H: S$$

Hence Euler's theorem verified.

C: $U = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$

Sol:

$$U = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \quad \text{--- (1)}$$

By checking homogeneous function multiply with 't'

$$= \frac{(tx)^{1/4} + (ty)^{1/4}}{(tx)^{1/5} + (ty)^{1/5}}$$

$$= \frac{t^{1/4} (x^{1/4} + y^{1/4})}{t^{1/5} (x^{1/5} + y^{1/5})}$$

$$= t^{1/20} \cdot \left(\frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \right)$$

$$\therefore \frac{t^{1/4} \cdot t^{1/5}}{t^{1/5} \cdot t^{1/4}} = t^{1/20}$$

This show "U" is homogenous function with degree " $\frac{1}{20}$ ". So, Euler's theorem is:

$$\Rightarrow x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = \frac{1}{20} U \quad \text{--- (1)}$$

Partial diff (1) w.r.t "x"

$$\begin{aligned} \frac{\partial U}{\partial x} &= \frac{\partial}{\partial x} \left[\frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \right] \\ &= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{\partial}{\partial x} (x^{1/4} + y^{1/4}) - (x^{1/4} + y^{1/4}) \cdot \frac{\partial}{\partial x} (x^{1/5} + y^{1/5})}{(x^{1/5} + y^{1/5})^2} \\ &= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{1}{4} x^{-3/4} - (x^{1/4} + y^{1/4}) \cdot \frac{1}{5} x^{-4/5}}{(x^{1/5} + y^{1/5})^2} \\ \frac{\partial U}{\partial x} &= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{1}{4} x^{-3/4} - (x^{1/4} + y^{1/4}) \cdot \frac{1}{5} x^{-4/5}}{(x^{1/5} + y^{1/5})^2} \end{aligned}$$

Partial diff (2) w.r.t "y"

$$\begin{aligned} \frac{\partial U}{\partial y} &= \frac{\partial}{\partial y} \left[\frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \right] \\ &= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{\partial}{\partial y} (x^{1/4} + y^{1/4}) - (x^{1/4} + y^{1/4}) \cdot \frac{\partial}{\partial y} (x^{1/5} + y^{1/5})}{(x^{1/5} + y^{1/5})^2} \\ &= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{1}{4} y^{-3/4} - (x^{1/4} + y^{1/4}) \cdot \frac{1}{5} y^{-4/5}}{(x^{1/5} + y^{1/5})^2} \end{aligned}$$

$$= \frac{(x^{1/5} + y^{1/5}) \cdot \frac{1}{4} y^{-3/4} - (x^{1/4} + y^{1/4}) \cdot \frac{1}{5} x^{-4/5}}{(x^{1/5} + y^{1/5})^2}$$

Put in eq ①

$$x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} = \frac{1}{20} U$$

$$\frac{x(x^{1/5} + y^{1/5}) \left[\frac{1}{4} x^{-3/4} \right] - (x^{1/4} + y^{1/4}) \left[\frac{1}{5} x^{-4/5} \right] + y(x^{1/5} + y^{1/5}) \left[\frac{1}{4} y^{-3/4} \right] - (x^{1/4} + y^{1/4}) \left[\frac{1}{5} y^{-4/5} \right]}{(x^{1/5} + y^{1/5})^2} = \frac{1}{20} U$$

$$\frac{x^{1/5} + y^{1/5}}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{4} x^{1/4} \right] - \frac{(x^{1/4} + y^{1/4})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{5} x^{1/5} \right] + \frac{(x^{1/5} + y^{1/5})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{4} y^{1/4} \right] - \frac{(x^{1/4} + y^{1/4})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{5} y^{1/5} \right] = \frac{1}{20} U$$

$$\frac{x^{1/5} + y^{1/5}}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{4} x^{1/4} \right] - \frac{(x^{1/4} + y^{1/4})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{5} x^{1/5} \right] + \frac{(x^{1/5} + y^{1/5})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{4} y^{1/4} \right] - \frac{(x^{1/4} + y^{1/4})}{(x^{1/5} + y^{1/5})^2} \left[\frac{1}{5} y^{1/5} \right] = \frac{1}{20} U$$

$$\frac{(x^{1/5} + y^{1/5}) \left[\frac{1}{4} x^{1/4} + \frac{1}{4} y^{1/4} \right] - \frac{1}{5} (x^{1/4} + y^{1/4}) (x^{1/5} + y^{1/5})}{(x^{1/5} + y^{1/5})^2} = \frac{1}{20} U$$

$$\frac{(x^{1/5} + y^{1/5}) \left[\frac{1}{4} - \frac{1}{5} \right]}{(x^{1/5} + y^{1/5})^2} = \frac{1}{20} U$$

$$= \frac{(x^{1/4} + y^{1/4}) \left[\frac{1}{4} - \frac{1}{5} \right]}{(x^{1/5} + y^{1/5})} = \frac{1}{20} U$$

$$\frac{(x^{1/4} + y^{1/4}) \left[\frac{1}{20} \right]}{(x^{1/5} + y^{1/5})} = \frac{1}{20} U$$

$$= \frac{1}{20} \cdot \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} = \frac{1}{20} U$$

$$\therefore U = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$$

$$- \frac{1}{20} U = - \frac{1}{20} U$$

$$L: H: S = R: H: S$$

So, Euler's Theorem Verified.

② we have given:-

$$U = b \left(\frac{y}{x} \right)$$

Sol:

$$U = b \left(\frac{y}{x} \right) \text{ --- ①}$$

Partial diff ① wr.t "x"

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x} \left[b \left(\frac{y}{x} \right) \right]$$

$$= b' \left(\frac{y}{x} \right) \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= b' \left(\frac{y}{x} \right) \cdot y \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$= b' \left(\frac{y}{x} \right) \cdot y \cdot \left(\frac{-1}{x^2} \right)$$

$$\frac{\partial U}{\partial x} = - \frac{y}{x^2} \cdot b' \left(\frac{y}{x} \right)$$

Partial diff ① w.r.t "y"

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left[b\left(\frac{y}{x}\right) \right]$$

$$= b'\left(\frac{y}{x}\right) \cdot \frac{\partial}{\partial y} \left(\frac{y}{x}\right)$$

$$= b'\left(\frac{y}{x}\right) \cdot \frac{1}{x} \cdot \frac{\partial}{\partial y} (y)$$

$$= b'\left(\frac{y}{x}\right) \cdot \frac{1}{x} \cdot (1)$$

$$\frac{\partial v}{\partial y} = \frac{1}{x} \cdot b'\left(\frac{y}{x}\right)$$

put in formula,

$$x \cdot \frac{\partial v}{\partial x} + y \cdot \frac{\partial v}{\partial y} = 0$$

Multiply with 'x' and 'y'

$$x \frac{\partial v}{\partial x} = x \left(\frac{y}{x^2} \cdot b\left(\frac{y}{x}\right) \right)$$

$$x \frac{\partial v}{\partial x} = \frac{-y}{x} \cdot b'\left(\frac{y}{x}\right) \quad \text{--- ①}$$

$$y \frac{\partial v}{\partial y} = y \left(\frac{1}{x} \cdot b'\left(\frac{y}{x}\right) \right)$$

$$y \frac{\partial v}{\partial y} = \frac{y}{x} \cdot b'\left(\frac{y}{x}\right) \quad \text{--- ②}$$

Adding ① and ②

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = \frac{-y}{x} b'\left(\frac{y}{x}\right) + \frac{y}{x} b'\left(\frac{y}{x}\right)$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0$$

Its showed.

③

$$U = xy \cdot b\left(\frac{x}{y}\right)$$

Sol:

$$u = xy \cdot b\left(\frac{x}{y}\right) = x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \partial u$$

Partial diff w.r.t to "x"

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[x \cdot y \cdot b\left(\frac{x}{y}\right) \right]$$

$$= y \cdot \frac{\partial}{\partial x} \left[x \cdot b\left(\frac{x}{y}\right) \right]$$

$$= y \left[x \cdot b'\left(\frac{x}{y}\right) \cdot \frac{\partial}{\partial x} \left(\frac{x}{y}\right) + b\left(\frac{x}{y}\right) \cdot \frac{\partial}{\partial x} (x) \right]$$

$$= y \left[x \cdot b'\left(\frac{x}{y}\right) \cdot \frac{1}{y} \cdot \frac{\partial}{\partial x} + b\left(\frac{x}{y}\right) \right]$$

$$= xy \cdot b'\left(\frac{x}{y}\right) \cdot \frac{1}{y} + y \cdot b\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial x} = x \cdot b'\left(\frac{x}{y}\right) + y \cdot b\left(\frac{x}{y}\right)$$

Partial diff w.r.t "y"

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left[x \cdot y \cdot b\left(\frac{x}{y}\right) \right]$$

$$= x \cdot \frac{\partial}{\partial y} \left[y \cdot b\left(\frac{x}{y}\right) \right]$$

$$\begin{aligned}
 &= x \cdot \left[y \cdot b'\left(\frac{x}{y}\right) \cdot \frac{\partial}{\partial y} \left(\frac{x}{y}\right) + b\left(\frac{x}{y}\right) \cdot \frac{\partial}{\partial y} (y) \right] \\
 &= x \left[y \cdot b'\left(\frac{x}{y}\right) x \cdot \frac{\partial}{\partial y} \left(\frac{1}{y}\right) + b\left(\frac{x}{y}\right) \right] \\
 &= x \left[y \cdot b'\left(\frac{x}{y}\right) x \left(-\frac{1}{y^2}\right) + b\left(\frac{x}{y}\right) \right]
 \end{aligned}$$

$$\frac{\partial U}{\partial y} = -\frac{x^2}{y} b'\left(\frac{x}{y}\right) + x b\left(\frac{x}{y}\right)$$

Multiply x and y with both above equation

With x value:

$$x \frac{\partial U}{\partial x} = x \cdot b'\left(\frac{x}{y}\right) + y \cdot b\left(\frac{x}{y}\right)$$

$$x \frac{\partial U}{\partial x} = x \left[x \cdot b'\left(\frac{x}{y}\right) + y \cdot b\left(\frac{x}{y}\right) \right]$$

$$x \frac{\partial U}{\partial x} = x^2 b'\left(\frac{x}{y}\right) + xy b\left(\frac{x}{y}\right) \quad \text{--- ①}$$

With y value:

$$\frac{\partial U}{\partial y} = -\frac{x^2}{y} b'\left(\frac{x}{y}\right) + x b\left(\frac{x}{y}\right)$$

$$y \frac{\partial U}{\partial y} = y \left[-\frac{x^2}{y} b'\left(\frac{x}{y}\right) + x b\left(\frac{x}{y}\right) \right]$$

$$= y \cdot -\frac{x^2}{y} b'\left(\frac{x}{y}\right) + xy b\left(\frac{x}{y}\right)$$

$$= -x^2 b'\left(\frac{x}{y}\right) + xy b\left(\frac{x}{y}\right) \quad \text{--- ②}$$

Adding ① and ②

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = x^2 \cancel{b'(\frac{x}{y})} + xy b(\frac{x}{y}) - x^2 \cancel{b'(\frac{x}{y})} + xy b(\frac{x}{y})$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 2xy b\left(\frac{x}{y}\right)$$

$$\because xy b\left(\frac{x}{y}\right) = v$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 2v$$

Hence proved:

④ $z = \tan^{-1} \left[\frac{y}{x} \right]$ verify $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Sol:

$$z = \tan^{-1} \left[\frac{y}{x} \right] \text{ --- ①}$$

Partial diff ① w.r.t "x"

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} \left[\tan^{-1} \left(\frac{y}{x} \right) \right]$$

$$= \frac{1}{1 + \left(\frac{y}{x} \right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right) \quad \because \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2} \cdot \frac{d}{dx} (x)$$

$$= \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot y \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$= \frac{x^2}{x^2+y^2} \cdot y \left(\frac{-1}{x^2} \right)$$

$$\frac{\partial z}{\partial x} = \frac{-y}{x^2+y^2}$$

Again diff w.r.t "x"

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left[\frac{-y}{x^2+y^2} \right]$$

$$= -y \cdot \frac{\partial}{\partial x} \left[\frac{1}{x^2+y^2} \right]$$

$$= -y \cdot \frac{\partial}{\partial x} (x^2+y^2)^{-1}$$

$$= -y \cdot -1 (x^2+y^2)^{-1-1} \cdot \partial x$$

$$= -y \cdot (x^2+y^2)^{-2} \cdot \partial x$$

$$\frac{\partial^2 z}{\partial^2 x} = \frac{\partial xy}{(x^2+y^2)^2} \quad \text{--- (2)}$$

Partial diff @ w.r.t "y"

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} \left[\tan^{-1} \left(\frac{y}{x} \right) \right]$$

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial y} \left[\frac{y}{x} \right]$$

$$= \frac{1}{x^2+y^2} \cdot \frac{1}{x} \quad (1)$$

$$= \frac{x^x}{x^2+y^2} \cdot \frac{1}{x}$$

$$\frac{\partial z}{\partial y} = \frac{x}{x^2+y^2} \quad \text{--- (2)}$$

Again diff w.r.t y

$$\frac{\partial^2 z}{\partial^2 y} = \frac{\partial}{\partial y} \left[\frac{x}{x^2+y^2} \right]$$

$$= x \cdot \frac{\partial}{\partial y} (x^2+y^2)^{-1}$$

$$= x \cdot -1 \cdot (x^2+y^2)^{-1-1} \cdot \partial y$$

$$= x \cdot (x^2+y^2)^{-2} \cdot \partial y$$

$$\frac{\partial^2 z}{\partial^2 y} = \frac{-2xy}{(x^2+y^2)^2} \quad \text{--- (3)}$$

~~Multiply~~ Adding (2) and (3)

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial^2 y} = \left(\frac{2xy}{(x^2+y^2)^2} + \frac{(-2xy)}{(x^2+y^2)^2} \right)$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial^2 y} = 0$$

Hence proved.

⑤ $u = \sin^{-1} \left[\frac{x^2 + y^2}{x+y} \right]$

verify $x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \tan u$

Sol: By substitution method

let $u = \sin^{-1} \left[\frac{x^2 + y^2}{x+y} \right]$

$z = u = \sin^{-1} \left[\frac{x^2 + y^2}{x+y} \right]$

$z = \sin u = \frac{x^2 + y^2}{x+y} \quad \text{--- (1)}$

$z = \frac{x^2 + y^2}{x+y}$

Multiply with "t"

$z = \frac{(tx)^2 + (ty)^2}{tx + ty}$

$z = t \cdot \left[\frac{x^2 + y^2}{x+y} \right] \quad \text{--- (2)}$

Hence z is Homogenous function with $n=1$ so Euler's theorem is:

$x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y} = (1)z$

from (1) put $z = \sin u$

$x \cdot \frac{\partial}{\partial x} \sin u + y \cdot \frac{\partial}{\partial y} \sin u = \sin u$

$x \cdot \cos u \cdot \frac{\partial u}{\partial x} + y \cdot \cos u \cdot \frac{\partial u}{\partial y} = \sin u$

Divide both side by $\cos u$:

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \frac{\sin u}{\cos u}$$

$$\text{i.e. } \frac{\sin}{\cos} = \tan$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \tan u$$

Hence proved.

⑥

$$U = \sin^{-1} \left[\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right]$$

verify

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 0$$

Sol:

By substitution method.

$$U = \sin^{-1} \left[\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right]$$

$$Z = U = \sin^{-1} \left[\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right]$$

$$Z = \sin U = \left[\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right] \quad \text{--- (1)}$$

$$Z = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$$

Multiply or Divide by "t"

$$= \frac{\sqrt{tx} - \sqrt{ty}}{\sqrt{tx} + \sqrt{ty}}$$

$$\text{e.g. } t^0$$

$$Z = t^0 \cdot \left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right)$$

Euler's theorem:

$$x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y} = 0$$

put $z = \sin u$ from (1)

$$x \cdot \frac{\partial (\sin u)}{\partial x} + y \cdot \frac{\partial (\sin u)}{\partial y} = 0$$

$$x \cdot \cos u \cdot \frac{\partial u}{\partial x} + y \cdot \cos u \cdot \frac{\partial u}{\partial y} = 0$$

Divided both side by $\cos$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \frac{0}{\cos}$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 0$$

(7)

$$u = \ln \left[\frac{x^2 + y^2}{x + y} \right]$$

$$\text{Verify } x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 1$$

Sol:

$$u = \ln \left[\frac{x^2 + y^2}{x + y} \right]$$

Partial diff w.r.t "x"

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[\ln \left[\frac{x^2 + y^2}{x + y} \right] \right]$$

$$= \frac{1}{\frac{x^2 + y^2}{x + y}} \cdot \frac{\partial}{\partial x} \left(\frac{x^2 + y^2}{x + y} \right)$$

$$= \frac{x+y}{x^2+y^2} \cdot \frac{(x^2+y^2) \cdot \partial x - (x^2+y^2)(1)}{(x^2+y^2)^2}$$

$$\frac{\partial V}{\partial x} = \frac{\partial x (x+y) - (x^2+y^2)}{(x+y)(x^2+y^2)}$$

Partial diff w.r.t "y"

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y} \left[\ln \left(\frac{x^2+y^2}{x+y} \right) \right]$$

$$= \frac{1}{\frac{x^2+y^2}{x+y}} \cdot \frac{\partial}{\partial y} \left(\frac{x^2+y^2}{x+y} \right)$$

$$= \frac{x+y}{x^2+y^2} \cdot \frac{(x+y) \cdot \partial y - (x^2+y^2)(1)}{(x+y)^2}$$

$$\frac{\partial V}{\partial y} = \frac{\partial y (x+y) - (x^2+y^2)}{(x+y)(x^2+y^2)}$$

Multiply x and y with partial values

$$x \cdot \frac{\partial V}{\partial x} = x \left[\frac{\partial x (x+y) - (x^2+y^2)}{(x+y)(x^2+y^2)} \right]$$

$$x \cdot \frac{\partial V}{\partial x} = \frac{x^2 (x+y) - x(x^2+y^2)}{(x+y)(x^2+y^2)} \quad \text{--- ①}$$

$$y \cdot \frac{\partial V}{\partial y} = y \left[\frac{\partial y (x+y) - (x^2+y^2)}{(x+y)(x^2+y^2)} \right]$$

$$y \cdot \frac{\partial v}{\partial y} = \frac{\partial y^2(x+y) - y(x^2+y^2)}{(x+y)(x^2+y^2)} \quad \text{--- (2)}$$

Adding (1) and (2)

$$x \cdot \frac{\partial v}{\partial x} + y \cdot \frac{\partial v}{\partial y} =$$

$$\frac{\partial x^2(x+y) - x(x^2+y^2)}{(x+y)(x^2+y^2)} + \frac{\partial y^2(x+y) - y(x^2+y^2)}{(x+y)(x^2+y^2)}$$

$$= \frac{\partial x^2(x+y) - x(x^2+y^2)}{(x+y)(x^2+y^2)} + \frac{\partial y^2(x+y) - y(x^2+y^2)}{(x+y)(x^2+y^2)}$$

$$= \frac{2(x+y)[x^2+y^2] - 1(x^2+y^2)(x+y)}{(x+y)(x^2+y^2)}$$

$$= \frac{(x+y)[x^2+y^2]}{(x+y)(x^2+y^2)} = 1$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 1$$

Hence proved.

(8)

$$u = f(x, y)$$

$$\text{verify } x f_x + y f_y = n f \quad \text{--- (1)}$$

Sol:

$$u = f(x, y)$$

So, Euler's theorem is:

$$x f_x + y f_y = n f \quad \text{--- (2)}$$

Partially diff ① w.r.t "x"

$$x b_{xx} + (1) b_x + y b_{yx} + b_y(0) = n b_x$$

$$x b_{xx} + b_x + y b_{yx} = n b_x$$

Multiply with "x"

$$x^2 b_{xx} + x b_x + xy b_{yx} = n x b_x \quad \text{--- (2)}$$

$$b_x = b_{xx}$$

$$b_y = b_{yx}$$

Partially diff ① w.r.t "y":

$$x b_{xy} + b_x(0) + y b_{yy} + b_y(1) = n b_y$$

$$x b_{xy} + y b_{yy} + b_y = n b_y$$

$$xy b_{xy} + y^2 b_{yy} + y b_y = n y b_y \quad \text{--- (3)}$$

Adding (iii) and (ii)

$$x^2 b_{xx} + x b_x + xy b_{yx} + xy b_x + y^2 b_{yy} + y b_y = n x b_x + n y b_y$$

$$x^2 b_{xx} + y^2 b_{yy} + xy b_{yx} + xy b_{xy} + (x b_y + y b_x) = n(x b_x + y b_y)$$

$$x^2 b_{xx} + y^2 b_{yy} + xy b_{yx} + xy b_{xy} + n b = n(n b)$$

assume $b_{xy} = b_{yx}$

$$x^2 b_{xx} + y^2 b_{yy} + xy(b_{yx} + b_{xy}) = n(n b) - n b$$

$$x^2 b_{xx} + y^2 b_{yy} + xy(b_{xy} + b_{yx}) = n(n b) - n b$$

$$x^2 b_{xx} + y^2 b_{yy} + 2xy b_{xy} = n^2 b - n b$$

$$x^2 b_{xx} + y^2 b_{yy} + 2xy b_{xy} = n(n-1)b$$

⑨ $u = f(r) ; r = \sqrt{x^2 + y^2}$

prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r).$$

Sol:

$$u = f(r)$$

Partially diff "u" w.r.t "x"

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} f(r)$$

$$= f'(r) \cdot \frac{\partial r}{\partial x}$$

$$= f'(r) \cdot \frac{\partial}{\partial x} (x^2 + y^2)^{1/2}$$

$$= f'(r) \cdot \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot 2x$$

$$= f'(r) \cdot \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial u}{\partial x} = f'(r) \cdot \frac{x}{r}$$

Again partial diff w.r.t "x"

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left[f'(r) \cdot \frac{x}{r} \right]$$

$$= f'(r) \cdot \frac{\partial}{\partial x} \left(\frac{x}{r} \right) + f''(r) \cdot \frac{\partial}{\partial x} \left(\frac{x}{r} \right)$$

$$= f''(r) \left(\frac{x}{r} \right) \cdot \frac{x}{r} + f''(r) \cdot \frac{r(1) - x \frac{\partial}{\partial x}(r)}{r^2}$$

$$= b'(\cancel{r}) \cdot \frac{x^2}{r^2} + b''(r) \cdot \frac{r}{r^2} - \left(\frac{x^2}{r^2} \right)$$

$$= b''(r) \cdot \frac{x^2}{r^2} + b''(r) \cdot \frac{r^2 - x^2}{r^2}$$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ r^2 &= x^2 + y^2 \\ r^2 - x^2 &= y^2 \end{aligned}$$

$$= b''(r) \cdot \frac{x^2}{r^2} + b''(r) \cdot \frac{y^2}{r^2}$$

$$\frac{\partial^2 u}{\partial x^2} = b''(r) \cdot \frac{x^2}{r^2} + b''(r) \cdot \frac{y^2}{r^2} \quad \text{--- ①}$$

Similarly :

$$\frac{\partial^2 u}{\partial y^2} = b''(r) \cdot \frac{y^2}{r^2} + b'(r) \cdot \frac{x^2}{r^3} \quad \text{--- ②}$$

Adding ① and ②

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = b''(r) \cdot \frac{x^2}{r^2} + b'(r) \cdot \frac{y^2}{r^3} + b''(r) \cdot \frac{y^2}{r^2} + b'(r) \cdot \frac{x^2}{r^3}$$

$$= b''(r) \left[\frac{x^2}{r^2} + \frac{y^2}{r^2} \right] + b'(r) \left[\frac{y^2}{r^3} + \frac{x^2}{r^3} \right]$$

$$= b''(r) \left[\frac{x^2 + y^2}{r^2} \right] + b'(r) \left[\frac{y^2 + x^2}{r^3} \right]$$

$$= b''(r) \left[\frac{r^2}{r^2} \right] + b'(r) \left[\frac{r^2}{r^3} \right] \quad \because r^2 = x^2 + y^2$$

$$= b''(r) + \frac{1}{r} \cdot b'(r)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = b''(r) + \frac{1}{r} \cdot b'(r)$$

10

$$V = \rho^m ; \rho^2 = x^2 + y^2 + z^2$$

Verify that

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m+1)\rho^{m-2}$$

Sol:

$$V = \rho^m$$

Partial diff "V" wr.t "x"

$$\begin{aligned} \frac{\partial V}{\partial x} &= \frac{\partial}{\partial x} \rho^m \\ &= m \rho^{m-1} \frac{\partial}{\partial x} \rho \\ &= m \rho^{m-1} \frac{\partial \rho}{\partial x} \\ &= m \rho^{m-1} \cdot \rho^{-1} x \end{aligned}$$

$$\begin{aligned} \rho^2 &= x^2 + y^2 + z^2 \\ \frac{\partial \rho^2}{\partial x} &= 2x \\ \rho \frac{\partial \rho}{\partial x} &= x \\ \therefore \frac{\partial \rho}{\partial x} &= \frac{x}{\rho} \end{aligned}$$

$$\frac{\partial V}{\partial x} = m \rho^{m-2} \cdot x$$

Again diff partial wr.t "x"

$$\begin{aligned} \frac{\partial^2 V}{\partial x^2} &= m \cdot \frac{\partial}{\partial x} (x \cdot \rho^{m-2}) \\ &= m \left[x \cdot (m-2) \rho^{m-3} \cdot \frac{\partial \rho}{\partial x} + \rho^{m-2} \cdot (1) \right] \\ &= m \left[x \cdot (m-2) \rho^{m-3} \cdot \frac{x}{\rho} + \rho^{m-2} \right] \\ &= m x^2 (m-2) \rho^{m-4} + m \rho^{m-2} \end{aligned}$$

$$\frac{\partial^2 V}{\partial x^2} = mx^2(m-2)r^{m-4} + mr^{m-2} \quad \text{--- (1)}$$

Now,

Similarly:-

$$\frac{\partial^2 V}{\partial y^2} = my^2(m-2)r^{m-4} + mr^{m-2} \quad \text{--- (2)}$$

Similarly:-

$$\frac{\partial^2 V}{\partial z^2} = mz^2(m-2)r^{m-4} + mr^{m-2} \quad \text{--- (3)}$$

Adding (1), (2) and (3)

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = mx^2(m-2)r^{m-4} + mr^{m-2} +$$

$$my^2(m-2)r^{m-4} + mr^{m-2} + mz^2(m-2)r^{m-4} + mr^{m-2}$$

$$= 3mr^{m-2} + m(m-2)r^{m-4} [x^2 + y^2 + z^2]$$

$$= 3mr^{m-2} + m(m-2)r^{m-4} \cdot r^2 \quad \because x^2 + y^2 + z^2 = r^2$$

$$= 3mr^{m-2} + m(m-2)r^{m-2}$$

$$= (3m + m^2 - 2m)r^{m-2}$$

$$= (m^2 - m)r^{m-2}$$

$$= m(m+1)r^{m-2}$$

Hence verified.

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m+1)r^{m-2}$$